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Post-tectonic magnetizations from the Cordova gabbro, Ontario and Palaeozoic reactivation in the Grenville Province

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Summary. The Cordova gabbro of southern Ontario intrudes 1300 Myr old volcanic rocks of the Hastings Lowlands in the Grenville Structural Province. Three distinct vector magnetizations (A, B and C) have been isolated, using a combination of stable endpoints, subtracted vectors from orthogonal vector plots and converging remagnetization circles. The A magnetization, with mean direction $D = 294^\circ$ $I = -55.5^\circ$ ($k = 42$, $\alpha_{95} = 5.5^\circ$, $N = 18$ sites), is a high coercivity, high blocking temperature remanence recorded by 49 samples. The B magnetization was isolated in 33 samples and has a mean direction $D = 305.5^\circ$ $I = -1.5^\circ$ ($k = 24$, $\alpha_{95} = 9.5^\circ$, $N = 11$ sites). B has lower coercivities and blocking temperatures than A where the two are superimposed. The A and B palaeopoles, 151°E, 10.5°S ($dp = 6^\circ$, $dm = 8^\circ$) and 165.5°E, 24°N ($dp = 5^\circ$, $dm = 9.5^\circ$), fall on the Grenville Track around 900 and 820 Ma respectively. The A and B magnetizations thus date from uplift and cooling following the Grenvillian orogeny. The third magnetization, the C component, has been isolated in 23 samples. Its mean direction is $D = 180^\circ$ $I = 27.5^\circ$ ($k = 18$, $\alpha_{95} = 10.5^\circ$, $N = 12$ sites). The C is a low coercivity, low blocking temperature overprint of A and B. Its palaeopole, 102°E, 31°N ($dp = 6.5^\circ$, $dm = 12^\circ$), is unlike post-1300 Precambrian poles for cratonic North America but matches Silurian and late Ordovician poles. $^{40}Ar/^{39}Ar$ plateau ages of 446 and 447 Ma determined by Lopez-Martinez and York for plagioclases from one of the Cordova samples confirm this age assignment. The C magnetization therefore records a previously unrecognized mild thermal or hydrothermal event that occurred in Palaeozoic time, long after the Grenvillian orogeny.

Introduction

THE GRENVILLE PROVINCE AND THE GRENVILLIAN OROGENY

The Grenville Structural Province of the Canadian Precambrian Shield (Fig. 1) is an orogen of generally high-grade metamorphism that truncates structural trends of the less metamorphosed Southern, Superior, Churchill and Nain Provinces. In certain areas,

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highly metamorphosed equivalents of units in these more northerly provinces, such as the Huronian and Kaniapiskau supergroups, can be traced south into the Grenville Front Tectonic Zone, but these units tend to become unrecognizable more than a few tens of kilometres south of the Grenville Front. Grenvillian metamorphism was so profound that even in the Foreland Zone north of the Front, in apparently undisturbed and unmetamorphosed rocks, there is evidence in updated K/Ar ages of ≈ 1000 Ma reheating (Palmer, Merz & Hayatsu 1977).

The Archean basement gneisses and early and middle Proterozoic supracrustal rocks of the Grenville Province clearly experienced a profound tectonic/heating event whose aftermath was still being felt 1000 Ma. The probable mechanism for this Grenvillian orogeny is reactivation due to a Himalayan-type collision of Proterozoic continents (Dewey & Burke 1973), the Grenville being the deeply eroded analogue of modern-day Tibet. Estimates of 20–30 km of erosion in western Labrador and a crustal thickness during the Grenvillian orogeny of 60–70 km (Rivers 1983) are consistent with this picture and with the traditional view (e.g. Wynne-Edwards 1972) that the exposed Grenville has been eroded to catazonal depths.

On the other hand, in several widely separated parts of the Grenville, typical granulite or amphibolite terrains grade rapidly into areas of greenschist-facies metamorphism and/or unreset pre-metamorphic Rb/Sr or K/Ar ages. Examples of such lower-grade regions with a contrasting tectonometamorphic history are found in central Labrador (Emslie 1976; Rivers 1983), the Wakeham Group (Bourne 1978) in Quebec, and the Grenville Supergroup in the Hastings Lowlands of southern Ontario. The Cordova gabbro studied in this paper is located in this latter area.

THE HASTINGS LOWLANDS

The geology and metamorphism of the Hastings Lowlands (Fig. 1, inset) have been outlined by Lumbers (1964, 1969). The basal unit of the Grenville Supergroup is a thick (> 6 km) sequence of mostly mafic metavolcanics dated at 1310 ± 15 Ma (zircon U/Pb; Silver & Lumbers 1966). In the vicinity of the Cordova gabbro, the metavolcanics have been metamorphosed to the upper greenschist facies and form a steeply-dipping monoclinal sequence. Overlying carbonate and clastic metasediments are similarly deformed. Intruding the supracrustal sequence are a series of undeformed, lightly metamorphosed plutons. Oldest are epizonal granite and syenite stocks, intruded before the culination of regional metamorphism (Lumbers 1969), which have U/Pb zircon ages of 1250 ± 25 Ma (Silver & Lumbers 1966). These were followed by mesozonal trondhjemite stocks and mafic complexes, of which the Cordova body is one, that were intruded after the peak of regional metamorphism and give U/Pb zircon and $^{40}Ar/^{39}Ar$ hornblende ages of 1180–1200 Myr (S. B. Lumbers, cited in Symons 1978; Berger & York 1981). Contact metamorphic aureoles of many of these intrusions are well preserved.

Palaeomagnetic and $^{40}Ar/^{39}Ar$ geochronological studies indicate a prolonged post-metamorphic uplift and cooling history in this area and in the nearby Haliburton Highlands. The Thanet complex (Buchan 1978; Berger & York 1981) and the Cordova gabbro (Lopez-Martinez & York 1983) began cooling from above 600°C around 1200 Ma and reached 200°C around 900 Ma. The Haliburton intrusions (Buchan & Dunlop 1976; Berger, York & Dunlop 1979) had a parallel cooling history but the peak of metamorphism and onset of cooling lagged at least 100 Myr behind, in accord with the higher grade of metamorphism and implied deeper level of burial just west of the Hastings region. On a broader scale, differential uplift and cooling across the entire Grenville orogen is evident in K/Ar mica

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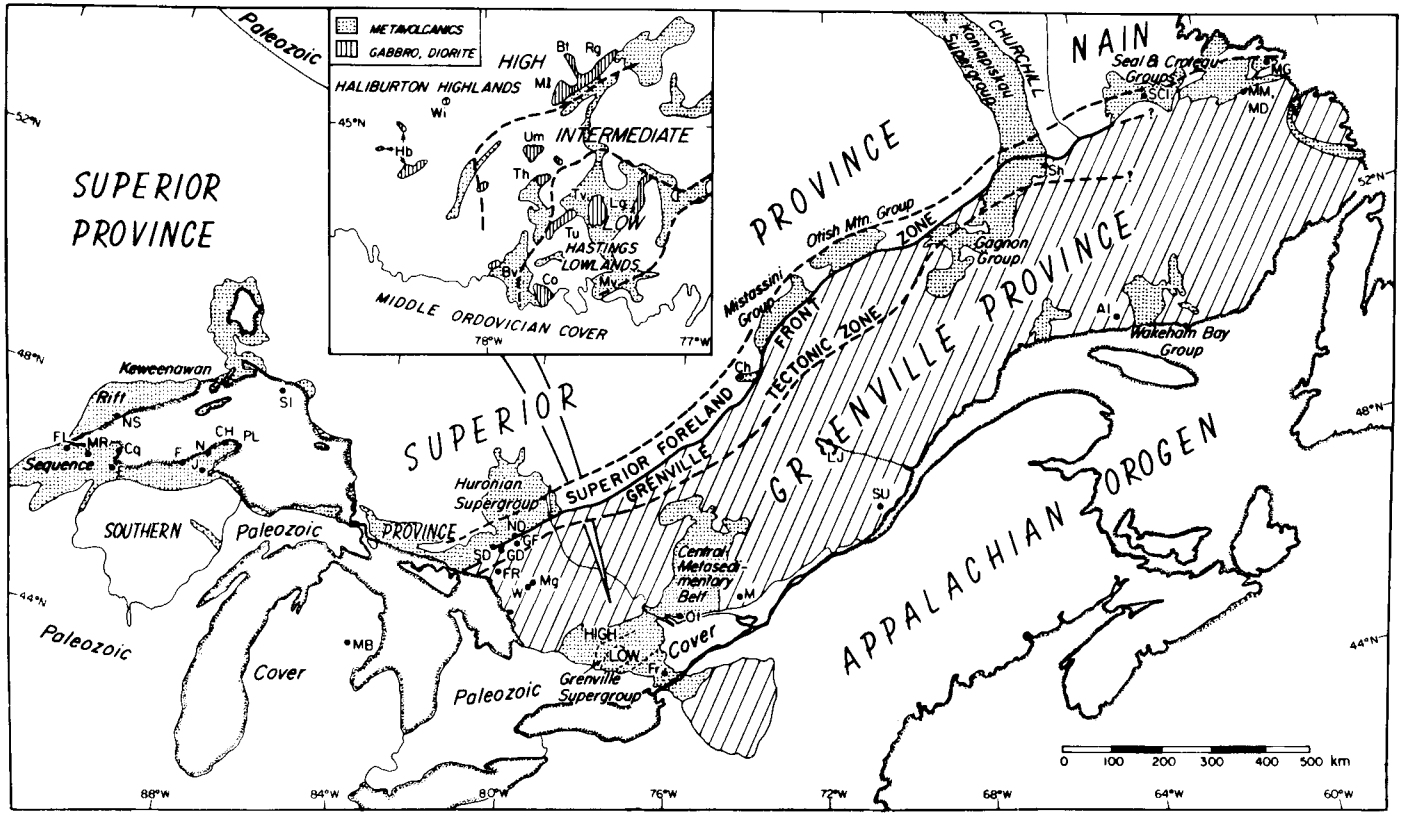


Figure 1. The Grenville Structural Province (hatched) of the Laurentian shield and (inset) the Hastings Lowlands of southern Ontario with its metamorphic zonation. Major supracrustal sequences within and outside the Grenville are shown stippled in the main map. Labels identify the sites of this study (Co) and other palaeomagnetic studies in the Grenville and Keweenawan. The same labels are used in Fig. 17(a, b) to identify palaeopoles from these studies and a key to the formations is given there. The limits of the Superior Foreland and Grenville Front Tectonic Zones follow Wynne-Edwards (1972).

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ages ('thermochrons' of Harper 1967 and Baer 1976) and palaeomagnetic poles (McWilliams & Dunlop 1978) that span the interval 1050–800 Ma (Berger *et al.* 1979).

The Grenvillian orogeny in the Hastings area thus had at least three phases. First there was deformation, largely monoclinal tilting, of thick volcanic and sedimentary sequences, followed by their burial to 10–15 km depths, perhaps by thrust-slice stacking (Rivers 1983), between 1300 and 1250 Ma. Then came intrusion of post-kinematic epizonal and mesozonal plutons around the time of deepest burial, between 1250 and 1200 Ma. Finally erosion, uplift and slow cooling occurred, either continuously or discontinuously, from 1200 to 900 Ma or later.

AN ALTERNATIVE VIEW OF THE GRENVILLIAN OROGENY

If uplift and cooling were continuous, the cooling rate must have been only $1-2^{\circ}C\ Myr^{-1}$, about an order of magnitude lower than estimated cooling rates in the Alps over the past few tens of million years (Baer 1981a). For this reason, and because of the concentration of 1100–1150 Myr old whole-rock Rb/Sr ages in the central part of the Grenville where the structural grain is NNE and oblique to the Grenville Front, Baer favours a compound cooling history with separate thermal events around 1150–1100 and 950–900 Ma.

The Central Metasedimentary Belt containing the Hastings region is one of the areas that should have felt the effects of a penetrative 1150–1100 Myr event, but the palaeomagnetic data do not bear this out. All palaeopoles so far reported seem to fall in the 1050–820 Myr interval and are very different from 1150–1100 Ma poles for rocks of the Lake Superior Keweenawan rift sequence (Fig. 1), which may be genetically related to the Grenvillian event or events (Donaldson & Irving 1972; Baer 1981b).

The palaeomagnetic observations do not rule out an 1150–1100 Ma event followed by reheating around 1050–1000 Ma to a temperature of about 650°C, which would be sufficient to erase all pre-existing magnetization. $^{40}Ar/^{39}Ar$ stepheating data for the Thanet complex (Berger & York 1981) are compatible with this compound model: one hornblende with an Ar closure temperature of 700°C gave an age of 1200 Myr whereas another with a closure temperature of 600°C gave an age of 1050 Myr, similar to ages of biotites that closed below 400°C. However hornblendes from the Cordova gabbro (Lopez-Martinez & York 1983) record simple cooling from 500°C around 1150 Ma to 350°C at 1050 Ma and rule out a reheating to 650°C just before 1050 Ma. For this reason, although the possibility of thermal pulses to moderate temperatures will be considered in what follows, a simple, very slow cooling history will be preferred at the outset.

Palaeomagnetic background

Grenvillian palaeomagnetic data and models for their interpretation have been reviewed by Symons (1978), Irving (1979), Roy & Robertson (1979), Dunlop (1981), Irving & McGlynn (1981) and Park & Emslie (1983). More specialized reviews of the palaeomagnetism and geochronology of the Hastings–Haliburton region have been given by Dunlop *et al.* (1980), Dunlop & Schutts (1983) and York (1984). Early studies were aimed at testing two different views of the 'Tibetan model' of the Grenville orogen. In the first view, the Grenvillian analogue of the Himalayan collision zone would lie near the southern edge of the Grenville Front Tectonic Zone (Fig. 1) and much of the Grenville would be an exotic terrane, with a pre-collisional apparent polar wander path (APWP) different from that of the remainder of Laurentia. In the other view, 'Grenvillia' and 'Interior Laurentia' form part of a single plate and the collisional zone lies to the SE of both.

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The discovery of multivectorial magnetizations in the Haliburton intrusions (Buchan & Dunlop 1973, 1976), the Morin complex (Irving, Park & Emslie 1974) and the Whitestone anorthosite (Ueno, Irving & McNutt 1974) lent impetus to the search for a pre-orogenic magnetic record. Individual samples recorded both an A magnetization with a palaeopole near Australia and a B magnetization with a palaeopole in the equatorial Pacific. The A magnetization was demonstrably older than the B, and the A-B sequence seemed to mark a convergent juncture with the Logan Loop or Keweenawan Track of the Laurentian APWP.

McWilliams & Dunlop (1978) hypothesized, however, that the A and B poles form part of a post-orogenic Grenville Track that is younger than any part of the Keweenawan Track. This conjecture was confirmed when the A and B poles were dated by $^{40}Ar/^{39}Ar$ thermochronometry (Berger *et al.* 1979; Berger & York 1981) and were shown to be about 1000–950 and 850–800 Ma respectively.

The Hastings Lowlands is one of the few parts of the Grenville from which pre-orogenic magnetizations might possibly be recovered. The discovery of primary K/Ar ages of 1150–1200 Myr for hornblendes from the Thanet and Cordova gabbros (Berger & York 1981; Lopez-Martinez & York 1983) intensified the search. Of course, whether a pre-metamorphic magnetization in either of these intrusions would be strictly pre-orogenic depends on whether the Grenvillian collision coincided with the 1150–1100 Ma event proposed by Baer (1981a) or with the earlier deformational event which predated these intrusions.

When a novel C component of magnetization was discovered in the Cordova gabbro (Dunlop *et al.* 1980), it was suggested on the basis of preliminary results that it might represent a pre-orogenic magnetization, or at any rate a remanence older than any previously uncovered. The detailed results presented in this paper show that this speculation was wrong. The C magnetization is a low-temperature Silurian or late Ordovician overprint, dated at 445–450 Ma by $^{40}Ar/^{39}Ar$ on plagioclase (Lopez-Martinez & York 1983). This is the first palaeomagnetic evidence for a possible 'echo' in the Grenville of Acadian–Caledonian events in the Appalachian Orogen.

Field and laboratory procedures

One hundred and thirty block samples were collected from 23 sites in the Cordova gabbro (Fig. 2). Samples were oriented by magnetic compass and strike readings were checked 1–2 m above the outcrop. Two sites (14 and 21) yielded no useful results, but almost all samples from the other sites behaved stably. The contact is poorly exposed, being covered by the Crowe River on the SW, by Crowe Lake on the south and by swamp to the east and NE. Paragneisses near the contact at site 14 and mafic metavolcanics near site 20 were collected for a baked contact test, but these samples proved to be weakly and erratically magnetized. Sixteen drill cores from the gold and sulphide mineralization zone of the former Cordova Mine near site 15 were stably magnetized, although interpretation of these results is more problematic since the cores were unoriented in azimuth.

Five 2.5 cm cylindrical specimens were drilled in the laboratory from each block sample or mine drill core. A full programme of stepwise alternating field (AF) and thermal demagnetization was carried out on every sample. Normally three specimens were AF cleaned in 12 steps to 1000 Oe (0.1 T) using a Schonstedt single-axis demagnetizer and one or two specimens were thermally cleaned in 50°C steps to 500°C and then in smaller steps to 660 or 670°C using either a home-built non-inductive furnace or a Schonstedt shielded furnace. Magnetizations were measured with Schonstedt and Digico spinner magnetometers. Results were accepted if specimen magnetizations reached a stable endpoint during cleaning with

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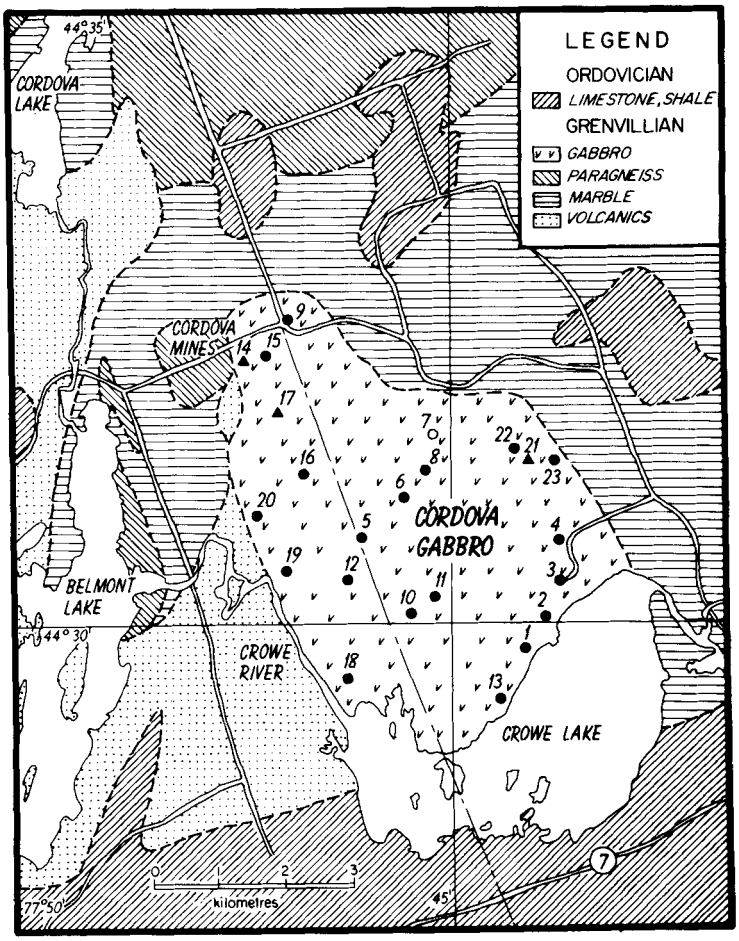


Figure 2. Palaeomagnetic sampling sites in the Cordova gabbro. Sites marked with a triangle gave few (17) or no (14, 21) stable results.

$< 20°$ between-specimen divergence or followed great-circle trajectories without reaching an endpoint.

Thin-section studies

Thin-sections of 23 samples were examined in transmitted light by L. D. Schutts. Average composition was 50–55 per cent clinopyroxene (augite), 40–45 per cent calcic plagioclase, <1 per cent biotite and the remainder coarse opaques, often rimmed by hornblende. The quantity of opaques averaged higher in samples with a C magnetization (10 sections examined) or C and an underlying A (four sections) than in samples with only an A magnetization (nine sections): 1–3 per cent of A sections, 2–3 per cent of C + A sections and 2–4 per cent of C sections.

SILICATE ALTERATION AND THE C MAGNETIZATION

There are also possibly significant differences in silicate alteration between C and A sections. In A sections, plagioclase is only slightly to moderately sericitized and saussuritization is

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seldom noted. In C + A sections, sericitization is moderate to extensive and in C sections it is extensive in all cases. In both C + A and C sections, saussuritization, where noted, is severe. Plagioclase alteration to scapolite is rare in A sections but common in C sections. Pyroxenes in A sections are sometimes uralitized to a hornblende-hastingsite only along their edges or in fractures, but in C + A and C sections, uralitization is invariably extensive.

One possible model of the alterations (L. D. Schutts 1982, private communication) is that uralitization and sericitization are deuteric while saussurite and scapolite formed hydrothermally. Uralitization at an early stage is reasonable enough: amphiboles that formed in this way are invariably chloritized to some degree (see below) and must have formed prior to regional metamorphism. There is no clear-cut evidence for the time of sericite formation. The association of sericite, saussurite and scapolite with the C magnetization, which is likely a low-temperature post-metamorphic overprint, favours a late hydrothermal origin for all three minerals. Furthermore alteration visibly associated with late-stage calcite-quartz veins was seen in about one-third of the C sections but not in A sections.

Biotites and uralites are lightly to extensively altered to chlorite, the degree of chloritization being unrelated to magnetization type. Several fine-grained calcite-albite-chlorite pods were also noted. These rocks must have partially re-equilibrated during regional greenschist metamorphism, in contradiction to the interpretation of Lumbers (1964, 1969) that the Hastings mafic stocks were intruded after the culmination of regional metamorphism.

MAGNETIC CARRIERS

In addition to the coarse discrete opaques, fine 'dusty' magnetite was observed in extensively uralitized pyroxenes. Plagioclases too are likely to contain fine-grained magnetite (e.g. Murthy, Evans & Gough 1971). These ultrafine magnetites, whose domain structure is

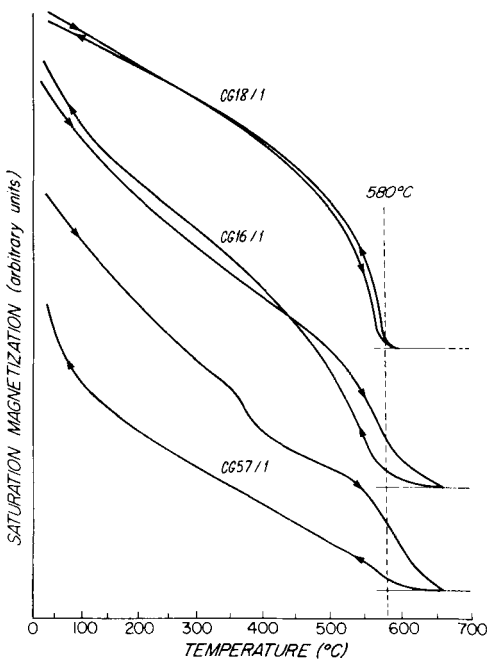


Figure 3. Representative strong-field thermomagnetic curves.

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likely to be single-domain (SD) or SD-like, are potentially a more important source of stable remanence than the coarse multidomain (MD) opaques.

Thermomagnetic curves

Thermomagnetic curves were measured in air using a recording Curie balance for 20 samples, of which 12 exhibited principally A or B natural remanent magnetizations (NRMs) and 8 principally C NRMs. The field applied was about 5 kOe (0.5 T). A few typical results are shown in Fig. 3.

Samples with C NRMs, like CG16 and 18, have practically reversible thermomagnetic behaviour and show a single Curie point near 580°C, indicating titanium-free magnetite as the only important magnetic phase. Although opaques are abundant in thin section, they tend to be altered and these samples have relatively low saturation magnetizations. The strong magnetite curve of CG18 is exceptional. Curves like that of CG16, with marked low-temperature paramagnetic curvature, are typical.

The curve for CG57 is fairly typical of results from samples with A or B NRMs. The magnetite Curie point is poorly indicated; there may be hematite present blurring the transition. Cooling curves almost always fall well below heating curves, so that magnetite has been destroyed, presumably by oxidation. There is also an indication, sometimes vague but occasionally quite pronounced as in CG57, of a magnetic phase with a Curie point or inversion temperature of 300–350°C. This phase could be pyrrhotite, maghemite or titanomagnetite. It is destroyed by heating, since cooling curves never show a low-temperature inflection.

The A component of NRM

STABLE ENDPOINT DIRECTIONS

After AF and thermal cleaning, the NRMs of 49 samples from 18 sites reached stable endpoint directions with NW declinations and intermediate to steep upward inclinations (reverse or R polarity relative to the present Earth's field or PEF) or antipodal directions (normal or N polarity), as shown in Fig. 4. The ratio of R to N is about 4 to 1. In most cases, the A remanence was revealed only after removal of a magnetization in the PEF direction between 0 and 150 Oe or between 20 and 250–300°C. This soft PEF component is presumed to be a room temperature viscous overprint acquired in the Brunhes epoch.

A statistical analysis of site mean A magnetization directions appears in Table 1. The N mean is not exactly 180° reversed to the R mean. N is displaced about 10° towards R along a great circle passing through the PEF direction, probably indicating imperfect removal of PEF viscous remanence. The mean A direction for R and N sites combined is $D = 294°$, $I = -55.5°$, with $k = 42$ and $\alpha_{95} = 5.5°$.

MAGNETIZATION AND FIELD REVERSALS

Fig. 5 illustrates N → R (or occasionally R → N) reversals observed in individual specimens in the course of AF or thermal cleaning. These reversals are often less than 180°, again likely because of unresolved PEF components.

The reversals are of several sorts. In CG79/1 and 5, the A′ remanence is so soft that it may actually be a modern viscous remanence with an unusual easterly declination. In CG76/4, only R-polarity A remanence is removed up to 575°C, above which an underlying N-polarity A′ remanence is revealed. The A′ remanence may well be a hematite

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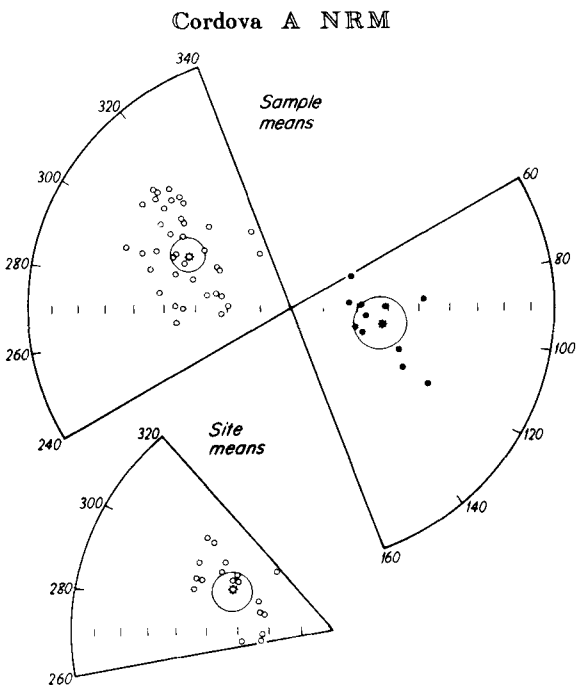


Figure 4. Equal-area plot of sample-mean and site-mean cleaned directions for the A magnetization. In this and later figures, open and closed symbols indicate upward and downward vectors respectively. Stars are grand means surrounded by their circles of 95 per cent confidence.

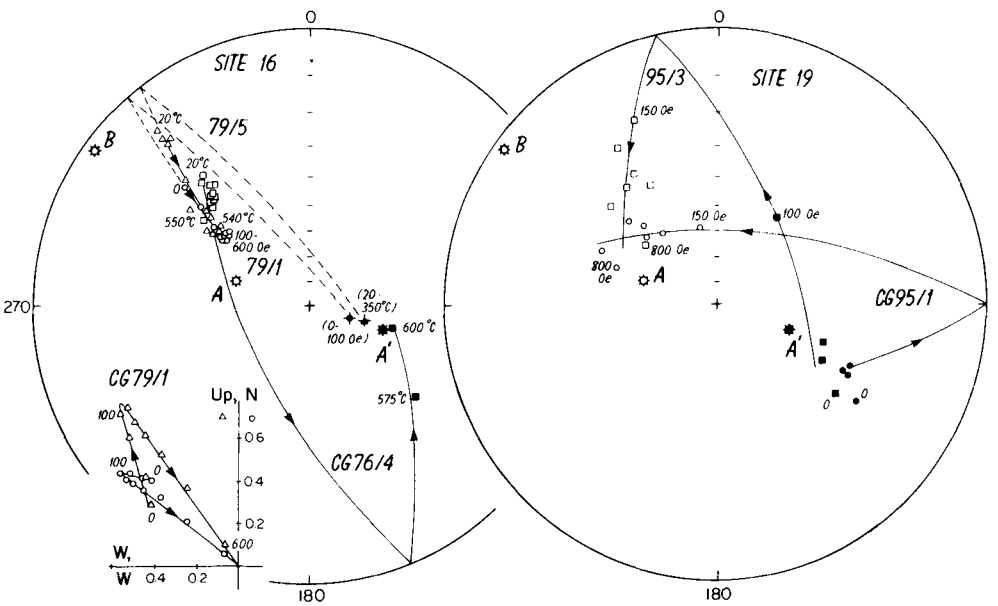


Figure 5. Stereographic and orthogonal vector projections of magnetization vectors in the course of stepwise AF and thermal demagnetization. On stereonets, symbols with a cross and labelled in parentheses are subtracted vectors between the AF or temperature steps indicated. In vector diagrams, triangles and circles denote vertical plane and horizontal plane projections respectively. Dashed lines are projected great circles joining subtracted vectors to measured data. The samples illustrated here exhibit a combination of reversed A and normal A (A′) magnetizations. The grand mean A, A′ and B directions are shown as stars.

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Table 1. Results of a 3-tier statistical analysis of Cordova gabbro A-component magnetization directions, based on AF and thermally cleaned endpoints and subtracted vectors. Four or five cleaned specimen magnetization vectors were averaged to obtain each sample mean (not listed); in turn, sample means were averaged to obtain the site means listed and site means were averaged to give grand means and their corresponding palaeopoles. N is the number of samples or sites averaged; D, I are declination and inclination of magnetization; k is the Fisher precision; and α_{95} is the radius of the 95 per cent confidence cone about the mean direction. N and R denote normal and reverse polarity respectively. N results were reversed for the combined averaging.

Site	N	D (°)	I (°)	k	α_{95} (°)
2	2	317	−64	57	
3	2	285	−66.5	1054	
4	4	287.5	−44	163	7.0
5	2	308	−39.5	78	
6	3	284	−67.5	151	10.0
8	1	299	−50		
9	2	292.5	−65	14	
11	2	264	−61	62	
12	4	300.5	−55	43	14.0
13	3	297.5	−56	36	20.5
15	2	262.5	−67.5	55	
16	5	308	−42	22	16.5
17	2	298	−55.5	209	
18	1	267.5	−68.5		
19	7	303.5	−49	34	10.5
20	2	298	−42	21	
22	3	292.5	−43.5	20	28.0
23	2	292.5	−44.5	10	
Mean, R samples	16 (38 samples)	298.0	−53.5	35	6.5
Mean, N samples	10 (11 samples)	101.0	61.5	36	7.5
Mean, unit weight to sites	18	294.0	−55.5	42	5.5
Palaeopole		151E	10.5S	(dp = 6°, dm = 8°)	

chemical remanence (CRM) self-reversed to the magnetite A remanence, rather than a thermoremanence (TRM) acquired during cooling above 575°C in an epoch of opposite field polarity. In CG95/1 and 3, magnetization vectors of different specimens follow great-circle paths of rather different azimuth, a hallmark of 180° reversals. The N-polarity NRM is rather soft and seems to reside in a different phase than the R-polarity NRM.

None of these examples taken individually is sufficient evidence of a field reversal but there are enough single-polarity N samples (Fig. 4) to make the case convincing. In bipolar (A + A′) samples, A′ almost always has low blocking temperatures. If A and A′ are partial TRMs, A records an earlier field than A′ and the field reversal must have been R → N. As we shall see later (in Fig. 17a), such a reversal fits the known polarity sequence for other Grenvillian data of similar direction and age.

BLOCKING TEMPERATURES AND COERCIVITIES

Decay of NRM intensity for single-component A or A′ samples is illustrated in Fig. 6(a). The majority of samples have relatively hard AF decay curves with SD-type curvature

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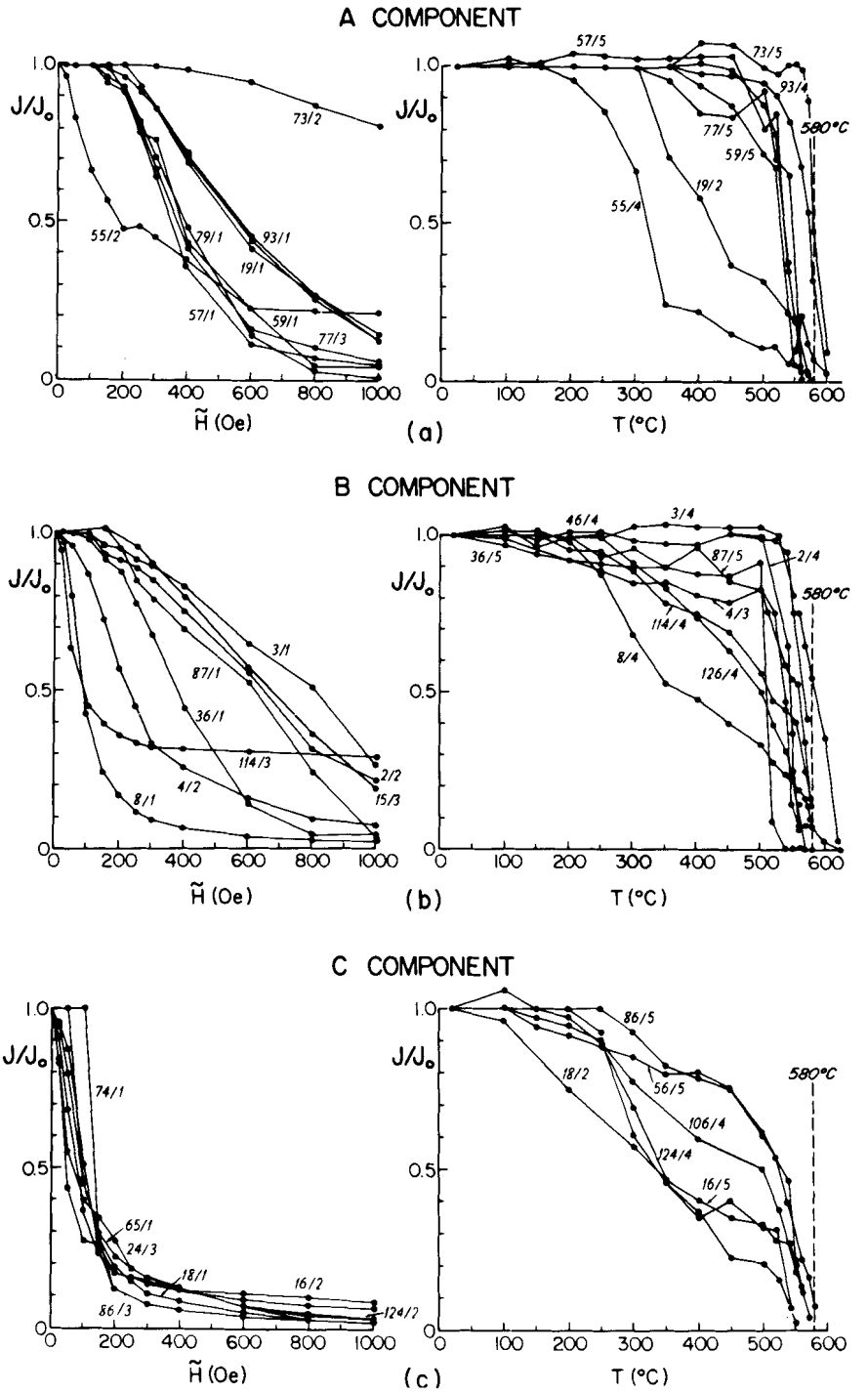


Figure 6. Intensity decay curves during AF and thermal demagnetization for samples with predominantly single-component A, B and C magnetization vectors [parts (a), (b) and (c) respectively]. The C coercivity and blocking temperature spectra are distinctly softer than the corresponding A and B spectra, which are rather similar to each other.

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(Bailey & Dunlop 1983) and median destructive fields (MDFs) of 350–550 Oe. The blocking temperature spectrum is discrete and concentrated between 520 and 570°C. The magnetic carrier is magnetite of SD or pseudo-SD size (<15 μm), probably inclusions in plagioclase and uralitized pyroxene.

A few samples like CG73 have very hard AF curves and blocking temperatures distributed well above 580°C. Hematite is an important NRM carrier in these samples. Other samples like CG55 have initially soft and irregular AF decay and a major (irreversible?) drop in intensity around 300°C. Maghemite, which inverts to weakly magnetic hematite at intermediate temperatures, is one candidate for the carrier of this remanence.

CONVERGING REMAGNETIZATION CIRCLES

The mean A direction is determined in Fig. 7 by an independent technique, the convergence of remagnetization circles (Halls 1976). Statistical parameters for the intersection technique

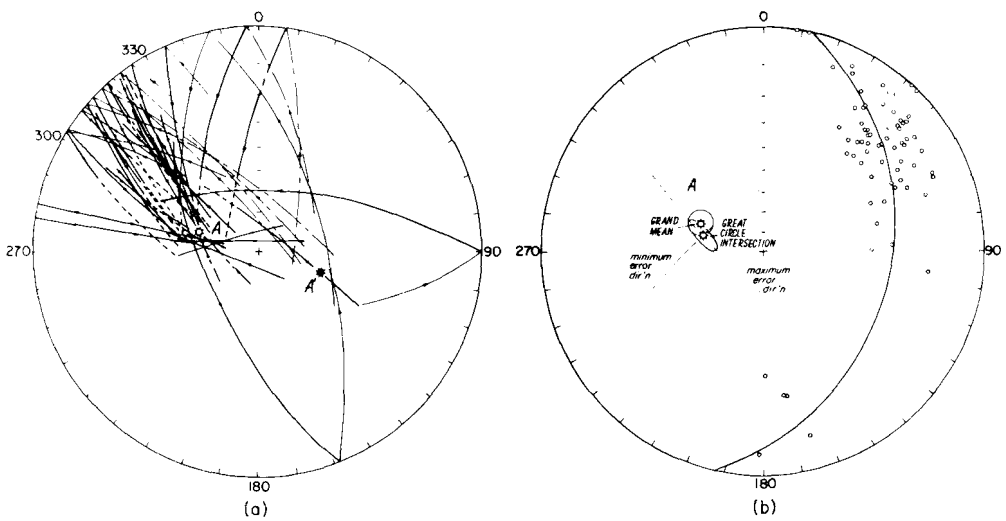


Figure 7. (a) The convergence of remagnetization circles like those of Fig. 5 on the A direction. Heavy and light lines are measured upper hemisphere and lower hemisphere great circles respectively; individual data points are not shown. Dot-dash lines are projected great circles anchored by subtracted vectors. (b) Normals to the planes represented by the great circles in (a) and other similar great circles, the best-fitting great circle through the normals, and the normal to the plane represented by this great circle, which is the best estimate of the convergence point in (a) (Halls 1976). The intersection point, with its oval of 95 per cent confidence (Bailey & Halls 1984), agrees well with the grand mean from Fig. 4.

Table 2. Cordova gabbro, analysis of remagnetization circles intersecting near the A and C component directions of Tables 1 and 4. The parameter k' corresponds to the Fisher k of Table 1 and the confidence oval semi-axes β_{95} and γ_{95} are analogous to α_{95}. The best intersection point is found by the method of Halls (1976) with statistical analysis following Bailey & Halls (1984).

	N	D (°)	I (°)	k'	β_{95} (°)	γ_{95} (°)
A, unit weight to specimen normals	70	285	−59	48	2.5	8.5
Palaeopole		154E	17.5S	(dp < 10°, dm < 13°)		
C, unit weight to specimen normals	78	172	16	13	6.0	9.5
Palaeopole		112E	37N	(dp < 5°, dm < 9.5°)		

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(Bailey & Halls 1984) are given in Table 2. The 95 per cent confidence oval about the mean intersection includes the grand mean of stable endpoints (Fig. 7b). These two estimates of the A direction are therefore indistinguishable at the 95 per cent confidence level.

The B component of magnetization

STABLE ENDPOINTS AND SUBTRACTED VECTORS

The B directions plotted in Fig. 8 were obtained by AF and thermal cleaning of 33 samples at 11 sites, either as stable endpoints or as resolved vectors from orthogonal vector plots of A + B resultants. Most B components of NRM have NW declinations and shallow positive or negative inclinations, usually between −20° and +25°. By comparison, A magnetizations

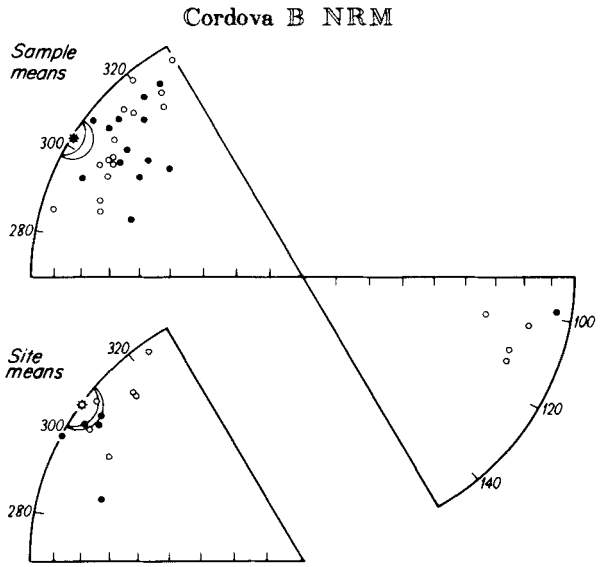


Figure 8. AF and thermally cleaned sample-mean and site-mean directions for the B magnetization vector.

Table 3. Results of a 3-tier analysis of Cordova gabbro B-component magnetization directions after AF and thermal cleaning. Only site means are listed. Symbols as in Table 1.

Site	*N*	*D* (°)	*I* (°)	*k*	α_{95} (°)
1	5	302.5	6.5	10	25.0
2	5	287	24.5	83	8.5
3	2	305.5	9.5	14.5	
4	1	298.5	−20.5		
6	2	315	−15	51.5	
8	4	297.5	0.5	15	24.5
10	4	303.5	11	25	18.5
18	2	308.5	−4.5	14	
20	4	324	−4.5	37	15.5
22	1	315	−13.5		
23	3	302	−8.5	9.5	42.0
Mean, unit weight to sites	11 (33 samples)	305.5	−1.5	24	9.5
Palaeopole		165.5 E	24 N	(dp = 5°, dm = 9.5°)	

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have inclinations of $-30°$ to $-70°$. There are only three borderline cases with inclinations between $-20°$ and $-30°$.

The B component is noticeably more scattered than the A. (It is for this reason that the remagnetization circles of Fig. 7(a) converge on A rather than on B.) The 95 per cent confidence cone about the mean B direction, $D = 305.5°$, $I = -1.5°$ (Table 3) is accordingly larger than for the A mean in Fig. 4: $\alpha_{95} = 9.5°$ compared to 5.5° for A. The Fisher precision k is 24, compared to 42 for A.

N-polarity B results are few in number and often have much softer coercivity and blocking temperature spectra than R-polarity results. It is therefore doubtful if any field reversal is recorded by the B component.

MULTIVECTORIAL DEMAGNETIZATION BEHAVIOUR

There are examples from virtually every stable site of bivectorial (A + B) or trivectorial (A + B + PEF) NRMs. Usually the A and B coercivity and blocking temperature spectra overlap, often to such an extent that no B direction can be determined from the demagnetization data.

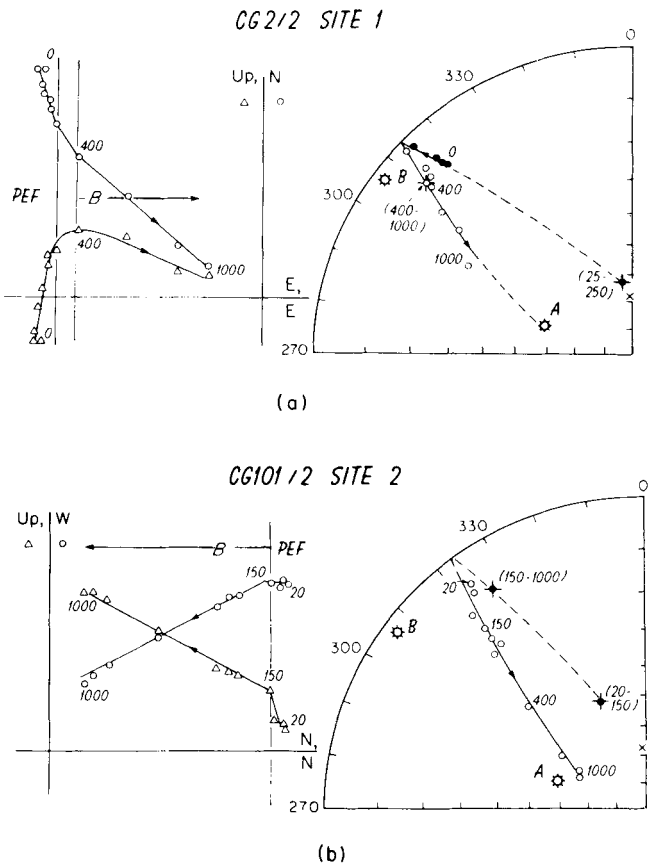


Figure 9. Orthogonal vector and stereographic projections of AF demagnetization data that reveal a superposition of PEF, B and A magnetizations. In these examples, the A NRM resides entirely in hematite, but more typically, magnetite is the main A carrier and A and B coercivity spectra overlap.

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Fig. 9 gives a few examples in which the B vector is cleanly resolved by AF demagnetization. The A magnetization in these samples is atypical: it resides entirely in hematite with coercivities >1000 Oe that do not overlap those of the B remanence. Thus between 250 Oe (which is sufficient to erase the PEF component) and 1000 Oe, only the B vector is attacked. In CG101/2, where the A and B vectors are of comparable magnitude, all B remanence is eliminated by 1000 Oe and the resultant vector approaches the A direction. In CG2/2, the resultant is midway between A and B at 1000 Oe, but this is a result of the small magnitude of A compared to B rather than of unusually high B coercitivites.

In general, A and B coercivity spectra are rather similar, although the B spectra are more variable (Fig. 6a, b). However, whenever A and B are superimposed in a sample, whatever the overlap of their coercivities, the resultant vector always migrates towards the A direction, showing that B is softer than A.

Similarly, in thermal demagnetization of multicomponent samples, the B vector is erased at lower temperatures than the A (Fig. 10). However, there are not many examples in orthogonal vector plots of non-overlapping A and B blocking temperature spectra. Single-component A and B blocking temperature spectra are also very similar (Fig. 6a, b). B therefore cannot be a simple thermal overprint of A.

The C component of magnetization

AF DEMAGNETIZATION

The C magnetization (Fig. 11) is a southward, shallow to intermediate positive vector whose demagnetization characteristics are quite different from those of the A and B NRMs, on which it is usually superimposed. The C vector is largely demagnetized in AFs < 200 Oe (Figs 6c, 11b), although a small fraction remains even at 1000 Oe.

Fig. 11 gives some particularly clear examples of multicomponent NRMs in which the C is prominent. Other examples could equally well have been chosen from 19 of the 23 sites. Although the C component is widespread in the Cordova body, frequently only one or two samples at a site exhibit it. Stable endpoints are rare and subtracted vectors are sometimes difficult to obtain because of overlapping coercivity spectra. The C data base thus relies heavily on the convergence of remagnetization circles like those of Fig. 11.

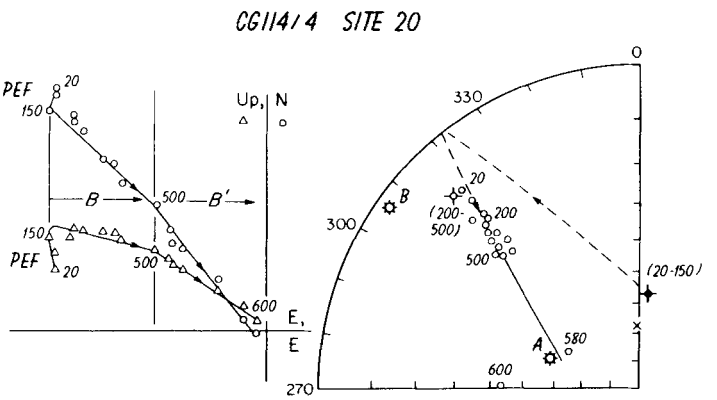


Figure 10. One example of a PEF + B + A multivector resolved in the course of thermal demagnetization. The components tend to be less cleanly separated in thermal cleaning than they were in AF cleaning (Fig. 9).

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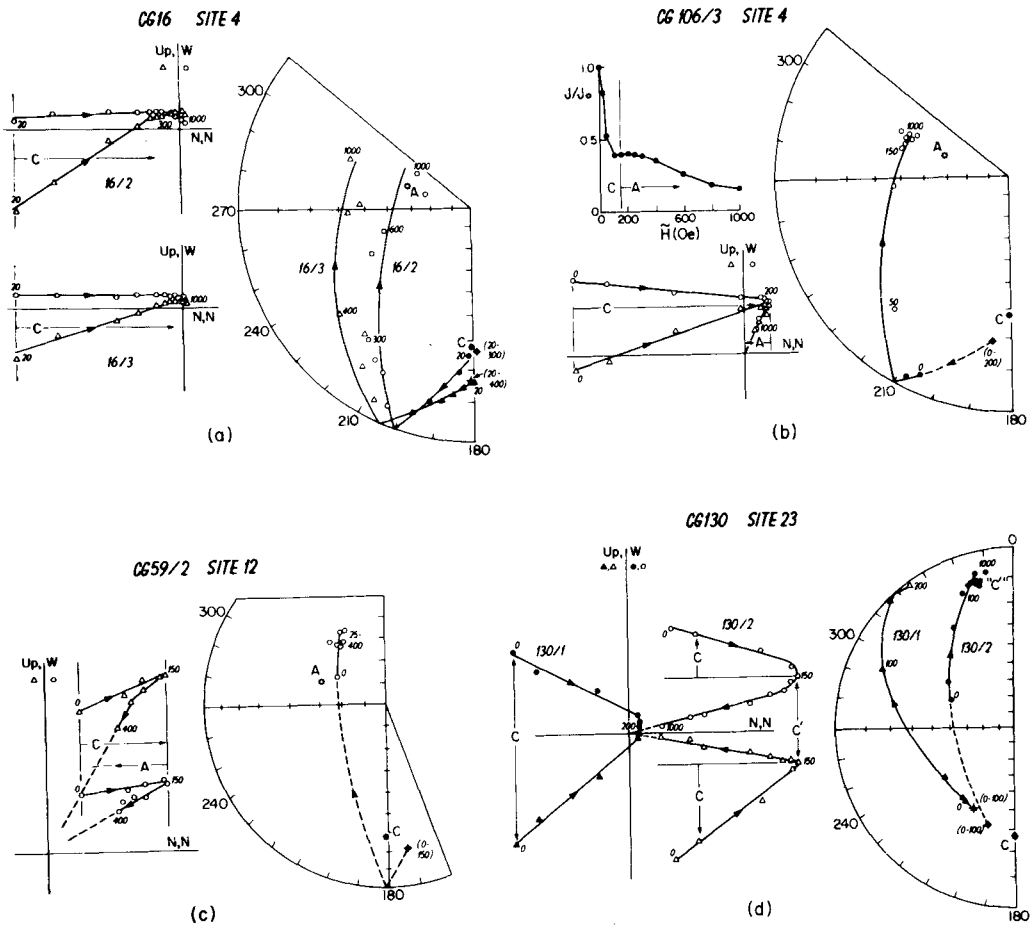


Figure 11. (a–c) Examples of the superposition of soft C and harder A magnetizations, revealed during AF cleaning. (d) A possible dual-polarity (C+C′) NRM vector in sample CG130. C and 'C′' are almost 40° from antiparallel in this sample.

CG16, 106 and 59 exhibit similar vector swings from the C direction at low fields toward the A direction at high fields. They differ mainly in the relative intensities of the A and C remanences. In CG16, the A component is small and hard (presumably hematite is responsible). The A direction is just reached at 1000 Oe, which is thus the upper limit of C coercivities in this sample, but the A vector is not demagnetized at all. In CG106, the A remanence is larger and softer. The A direction is approached by 150 Oe, and the A vector begins to be erased around 250 Oe, although almost 50 per cent of A remains at 1000 Oe. In CG59, A is larger than C. It is only in the orthogonal vector plot that the presence of C is clearly recognizable. The C and A coercivities are mainly <150 Oe and >400 Oe respectively.

CG130 (Fig. 11d) illustrates a less common type of behaviour: lower-hemisphere swings from the C direction towards an approximately reversed direction (C′) with no evidence of any A or B component. These 'reversals' are never 180° and there are few convincing C′ endpoints or subtracted vectors in the collection. A field reversal is probably not recorded.

The coercivity range of C is similar to that frequently displayed by PEF overprints.

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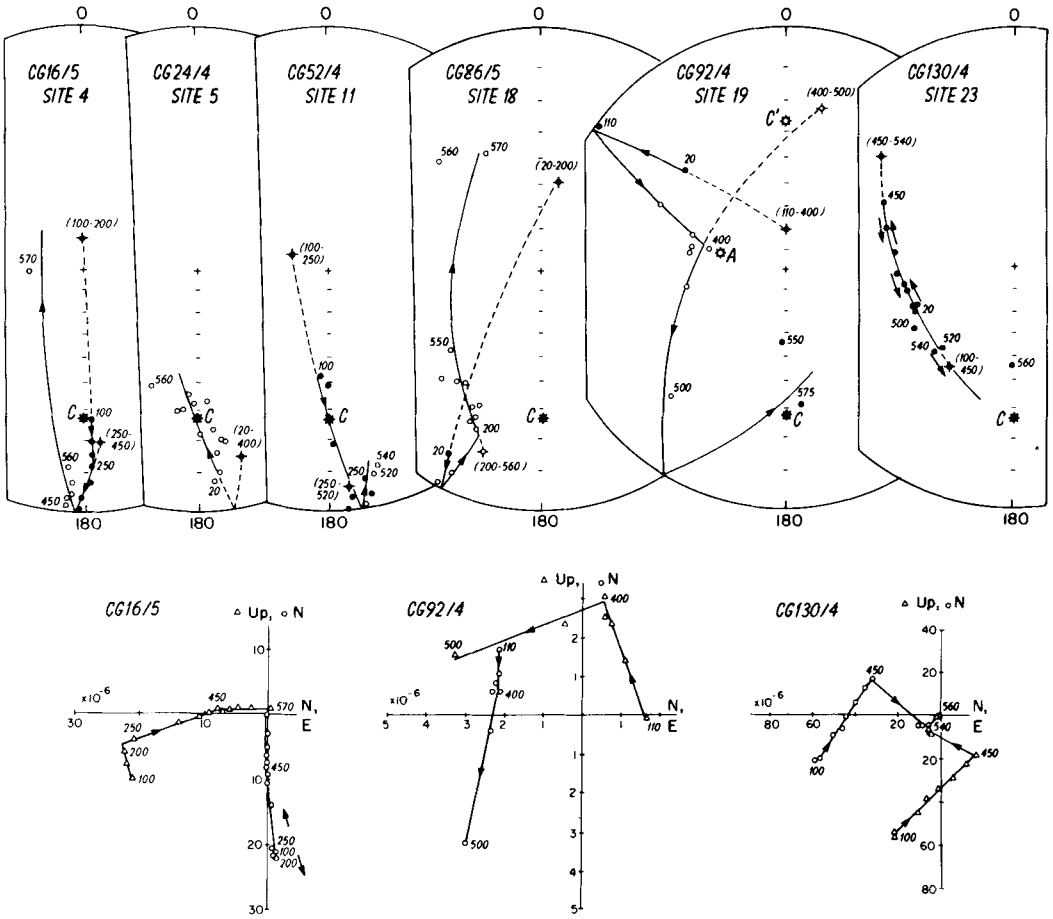


Figure 12. Examples of the C magnetization, usually overlain by PEF viscous remanence and underlain by an A component, revealed by thermal demagnetization.

However, C is definitely not a recent viscous magnetization, acquired during sampling or in the laboratory for example. There are numerous examples of stable C magnetizations with superimposed PEF overprints below 100 Oe or so. The similarity of C and PEF coercivity ranges may however explain the sporadic occurrence of C. Many of the C components originally present may have succumbed completely to PEF overprinting.

THERMAL DEMAGNETIZATION

The C component tends to have a larger proportion of low blocking temperatures (T_Bs) than A and B components (Fig. 6c). In most cases, however, the T_B spectrum extends to 580°C. C therefore cannot be entirely a partial thermoremanent magnetization (TRM) replacing low T_B fractions of A and B remanences.

The examples shown in Fig. 12 were chosen for their clean resolution of components into separate T_B ranges. CG16/5, 52/4 and 86/5 make the important point that PEF overprints occupy a T_B range, from 20 to 200 or 250°C, that is quite distinct from that of the C remanence. Above 250°C, C is either isolated as a stable endpoint (CG52/4) or, more

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typically, reveals an underlying A vector towards which the resultant swings up to 570°C (CG16/5, 24/4, 86/5).

CG92/4 and 130/4 are unusual but interesting examples. CG92/4 seems to have three coexisting ancient NRMs, A, C and C′, as well as a strong (and unusually hard) PEF overprint. The C′ vector is clearly displayed in the orthogonal vector plot between 400 and 500°C. The resultant vector swings from the A direction toward the C direction, which it reaches at 550°C, CG130, which displayed a possible C → C′, reversal on AF cleaning (Fig. 11d), swings only part of the way towards C′ on thermal cleaning, returning toward C with the removal of a (B?) component between 450 and 540°C. Such a component with a narrow range of T_Bs not obviously related to the Curie point of any common mineral is most unusual.

THE MEAN C DIRECTION

The rather sparse population of C stable endpoints and subtracted vectors is depicted in Fig. 13. There are relatively few N-polarity (C′) results and, as discussed above, it is questionable if these record a field reversal. There is more between-sample and between-site scatter than with A and B results (Figs 4 and 8), but there is no likelihood of confusing C with A or B data. The C has a mean direction (Table 4) $D = 180°$, $I = 27.5°$ for $N = 12$ sites, with $k = 18$ and $\alpha_{95} = 10.5°$.

Slightly better resolution of the mean direction is possible using the population of 78 specimen remagnetization circles that converge on C, some of which are shown in Fig. 14(a). The best-fitting great circle through all specimen normals (Fig. 14b) defines a convergence point at $D = 172°$, $I = 16°$ (Table 2) which is more tightly constrained than the grand mean of endpoints and subtracted vectors, although the two means do not differ significantly at 95 per cent confidence level. The N–S elongation of the error oval results from the large number of remagnetization circles that pass through the A direction.

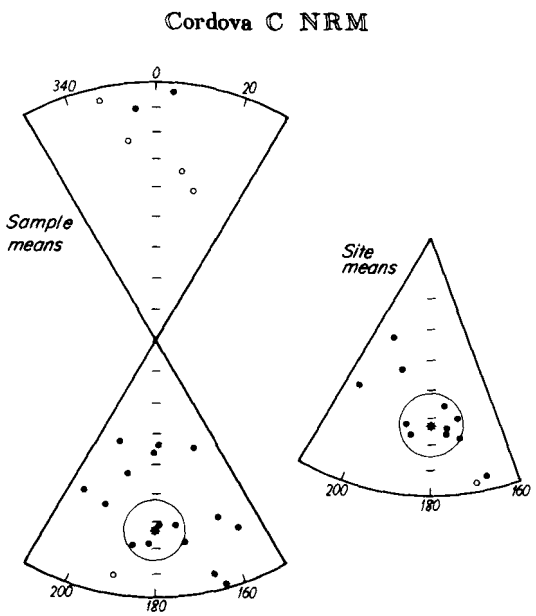


Figure 13. Sample-mean and site-mean C magnetizations, isolated as stable endpoints or as subtracted vectors in orthogonal projections during AF or thermal demagnetization.

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Table 4. Results of a 3-tier analysis of Cordova gabbro C-component magnetization directions after AF and thermal cleaning. Only site means are listed. Symbols as in Table 1.

Site	N	D (°)	I (°)	k	α_{95} (°)
2	2	172	29	14.5	
4	3	187.5	27.5	65	15.5
5	3	176	23.5	15.5	32.0
6	2	175.5	25.5	8.5	
8	1	199.5	55.5		
9	2	169.5	−4.5	45.5	
10	4	175.5	34	20	21.0
12	1	172	21.5		
13*	1	202.5	−2		
15*	1	183	−32		
18	2	186.5	23	8.5	
20	1	192	46.5		
22	1	167	5		
23	1	205.5	36		
Mean, unit weight to sites	12 (23 samples)	180.0	27.5	18	10.5
Palaeopole		102E	31N	(dp = 6.5°, dm = 12°)	

* Sites not included in grand mean (see text).

Multivectorial (A + B + C) NRMs

There are about 10 samples in the collection that display complex NRMs consisting of apparent A + B + C composites with PEF overprinting. Several examples are shown in Fig. 15. The PEF overprint is always removed by 200–250 Oe AF cleaning. Next to be attacked is an intermediate-coercivity NRM, which is sometimes B (CG48/1), sometimes A (CG92/3) and occasionally intermediate among A, B and C′ (CG122/3). In all cases,

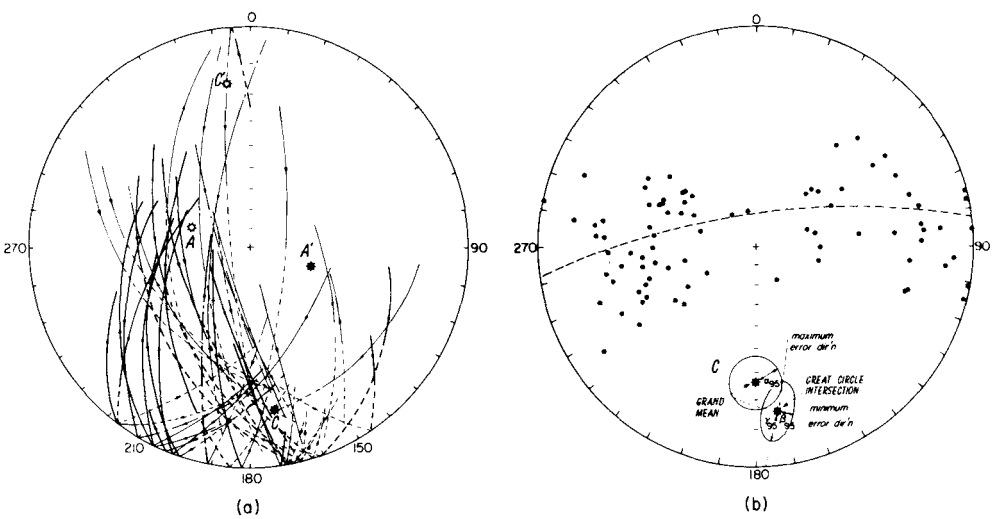


Figure 14. Remagnetization circles converging on the C direction, normals to the remagnetization planes, and the best estimate of the great-circle intersection point compared to the grand mean C direction from Fig. 13. Lines and symbols are as in Fig. 7, except that projected great circles are dashed.

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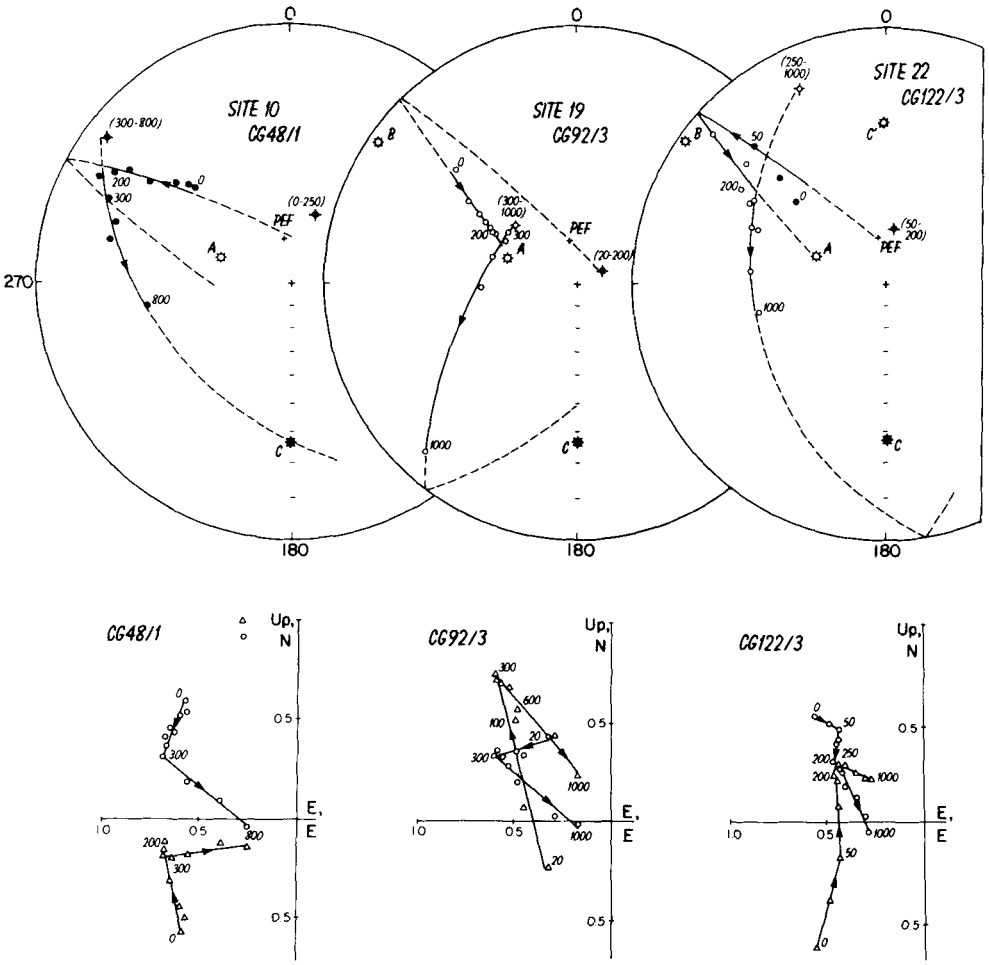


Figure 15. Examples of demagnetization data for samples with trivectorial (A + B + C) remanences. In all such samples, C is carried by hematite and is the hardest remanence.

the resultant vector swings *towards* the C direction at high fields, although C is not isolated by 1000 Oe. Thus in these samples, part of A and B and probably all of C are carried by hematite.

These results have a certain novelty but since they do not fit the general pattern of relative coercivities and blocking temperatures of A, B and C NRMs, they do little to advance our understanding of how and in what order these remanences were acquired. They will not be discussed further.

Cordova mines drillcores

There are a number of post-Ordovician sulphide vein and replacement deposits in the Hastings Lowlands whose mineralization ages, assuming their sources of radiogenic lead became isolated from a common reservoir around 1365–1000 Ma, are about 450 Myr old (Fletcher & Farquhar 1982, table 3). On the assumption that widespread mineralization might be one manifestiation of the event responsible for the C magnetization, 16 unoriented

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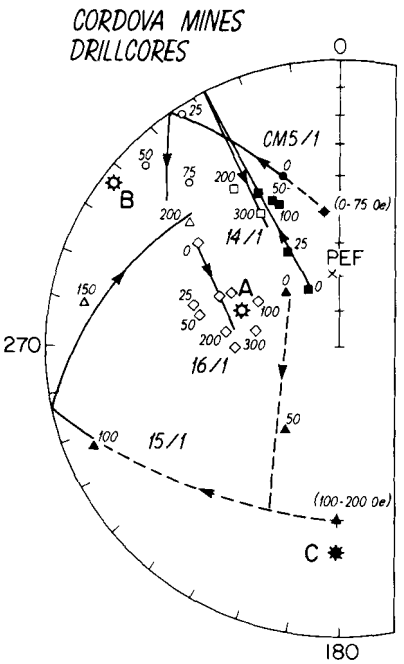


Figure 16. Selected AF demagnetization results for azimuthally unoriented drillcores from the Cordova Mines. Soft PEF overprints were used to determine north. PEF + B, B + A and PEF + C + A vectors are inferred from the data shown here.

(except for vertical) drillcores discarded from the core shed at the former Cordova Mines (Fig. 2) were examined to see if they displayed unusually prominent C magnetizations. It was anticipated that it would be possible to use soft PEF overprints to determine north and down for each core.

One specimen of each core was AF cleaned to 1000 Oe and a companion specimen of most cores was thermally cleaned to 620°C. A selection of the more interesting AF results is given in Fig. 16. NRM intensities dropped to low levels and directions tended to randomize at fields of 200–300 Oe. Five cores, including CM5/1 and 14/1 in Fig. 16, exhibited PEF to B or A swings before degenerating into noise. Two cores, of which 16/1 is an example, appeared to have B → A swings. CM15/1 and one other core had an initial very soft PEF overprint, a C component from 100 to 200 Oe, and an A or B component at higher fields. With the exception of one weakly magnetized sample, the remaining cores had similar soft intensity decay curves and were either directionally stable or followed great-circle swings, but the directions could not be confidently interpreted as combinations of A, B and C vectors, usually because there was no steep PEF component to fix the sample orientation.

Directional behaviour during thermal cleaning resembled AF behaviour for each core. PEF overprints were eliminated by 300°C and stable components had blocking temperatures extending to 560–580°C.

The Cordova Mines cores do not have particularly conspicuous C magnetizations and do not support the idea of a genetic relation between Palaeozoic sulphide mineralization and acquisition of the C magnetic overprint.

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Palaeopoles and apparent polar wander

Fig. 17(a–c) is a compilation of palaeopoles with Grenvillian affinities from within and outside of the Grenville Province. A few poles from formations just north of the Grenville Front have not been shown because of space limitations, but are plotted by Hyodo & Dunlop (1985). All published poles from the Grenville Province are shown in Fig. 17(a). They form a linked sequence from about 1020 to 820 Ma (the Grenville Track), followed by two probably later groups of poles, the Tu/Tu_A/SM grouping and the single pole Co_C of the present study.

Selected Keweenawan poles from the time window, 1150–1050 Ma, appear in Fig. 17(b). They link satisfactorily with the older end of the Grenville Track, except for primary poles MR_P, E_P and FL_P (Watts 1981), which fall well to the west of the Keweenawan Track among later Grenville poles. Fig. 17(c) summarizes Grenvillian or just pre-Grenvillian poles from Arizona, Montana, Wyoming, the Mackenzie Mountains and Baffin Island. Unfortunately, few of these poles are well dated radiogenically, but it is clear that Grenville-like poles are by no means confined to the Grenville Province.

Palaeopoles Co_A, Co_B and Co_C corresponding to the mean Cordova A, B and C magnetizations respectively appear in Tables 1, 3 and 4. These poles were calculated without corrections for tectonic rotations, since Co_A and Co_B resemble Grenvillian poles which are known to date from post-tectonic uplift and cooling and since Co_C is argued to be an even younger overprint. In Fig. 17(a), Co_A, Co_B and Co_C are compared explicitly to other Grenville Province palaeopoles.

Co_A does not fall in the southerly ≈1000 Ma group of A poles with Hb_A and Th_A of the Haliburton and Thanet intrusions of the Hastings–Haliburton region. Instead Co_A resembles more northerly poles about 900 Ma in age from the Wilberforce pyroxenite, the Umfraville gabbro and the Whitestone anorthosite, all of southern Ontario. However, Co_B does resemble Hb_B and Th_B. It is probably about 850–800 Myr old.

Fig. 17(d) compares Ordovician, Silurian and Devonian palaeopoles from cratonic North America with the Grenville Track. Co_C is virtually coincident, both in its position and in its 95 per cent confidence oval, with pole Bl of the Silurian Bloomsburg Formation (Roy, Opdyke & Irving 1967). It falls well west of most Ordovician poles and south and west of most Devonian poles, although it does resemble two Devonian poles (Do and PS1) that diverge from the main grouping. In view of the $^{40}Ar/^{39}Ar$ plagioclase age of about 445 Ma determined by Lopez–Martinez & York (1983) for the Cordova, an upper Ordovician–Silurian age for Co_C is most likely.

Discussion

A AND B NRMs: POST-GRENVILLIAN COOLING HISTORY

The Cordova A and B magnetizations are definitely not primary, and tell us nothing about the nature of the Grenvillian event or events. Like other Grenville A and B poles, they were acquired during post-tectonic uplift and cooling following the Grenvillian orogeny, in the general time interval 1000–800 Ma (Patchett & Bylund 1977; Berger *et al.* 1979; Dallmeyer & Sutter 1980). Judging by the paleopole positions, Co_A is about 900 Ma and Co_B is about 850–800 Myr old.

According to the regional cooling curve established by Berger & York (1981) and Lopez–Martinez & York (1983) from $^{40}Ar/^{39}Ar$ mineral ages and Ar closure temperatures, the Cordova and Thanet gabbros had cooled to around 500°C by 1150–1100 Ma and to around 300°C by 1000 Ma. If the Cordova A and B magnetizations were TRMs or partial

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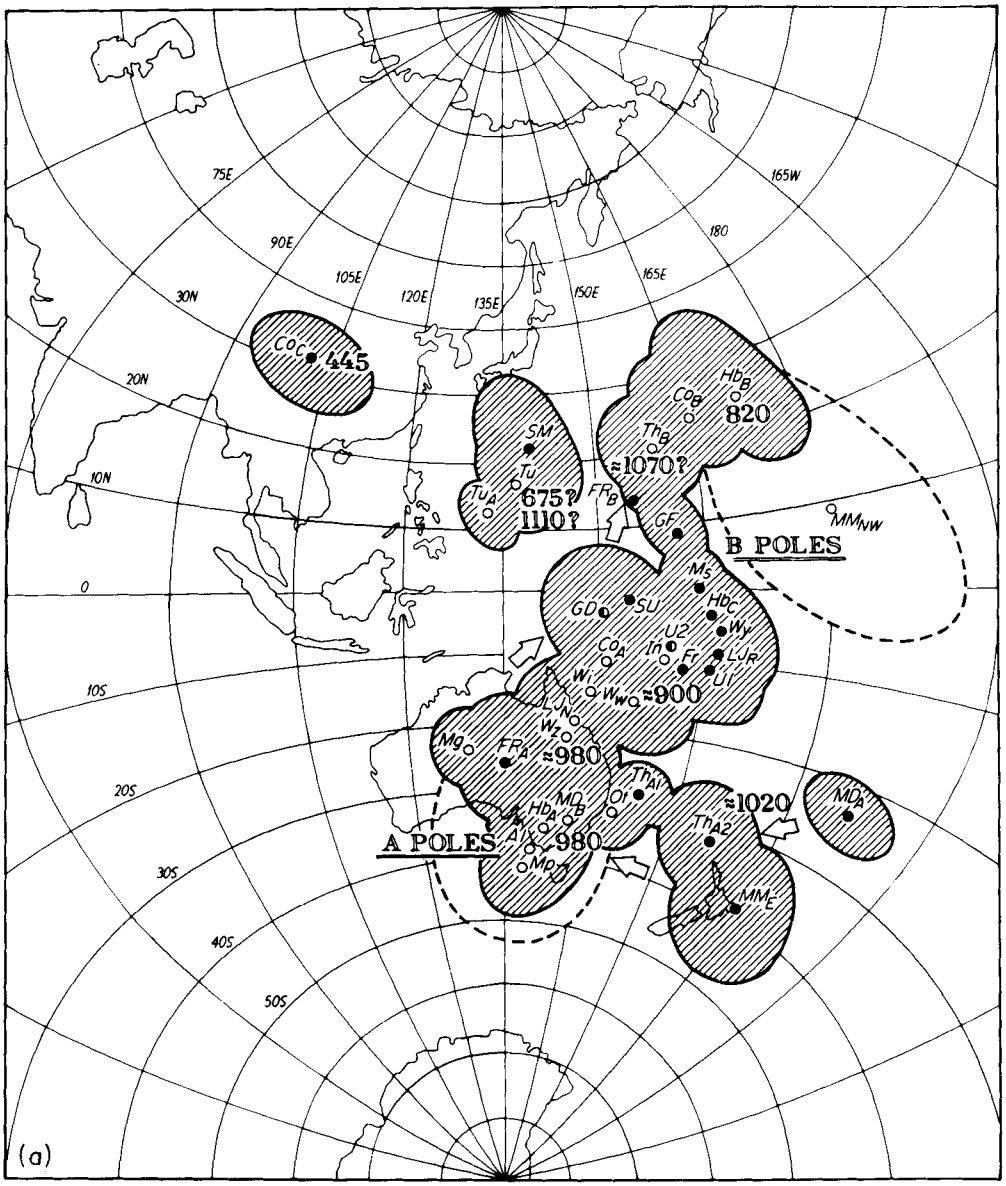


Figure 17. (a) Compilation of Grenville Province palaeopoles. Open, half-open and solid symbols denote reverse, mixed-polarity and normal results respectively. Ovals of 95 per cent confidence are hatched in, except for poles with large uncertainties, whose ovals are dashed. Numbers are $^{40}Ar/^{39}Ar$ ages in Myr. Recent dates by Reynolds (1984) of 1100–1000 Ma for MD_A and 1000 Ma for MD_B fit the general trend of ages. Unless otherwise indicated, all poles are referenced by Roy & Robertson (1979, table 1) or Buchan *et al.* (1983, table 3). Key to labels: MD_A, MD_B, Mealy Dykes (Park & Emslie 1983); MM_E, MM_{NW}, Mealy Mountains anorthosite; Th_{A1}, Th_{A2}, Th_B, Thanet gabbro complex; Ot, Ottawa intrusions; M_P, M_S, Morin anorthosite; Al, Allard Lake anorthosite; Hb_A, Hb_B, Hb_C (=Hb_1, Hb_3, Hb_2 of Buchan *et al.* 1983), Haliburton intrusions; Mg, Magnetawan metasediments; FR_A, French River anorthosite A component (Stupavsky & Symons 1982); W_W, W_Y, W_Z, Whitestone anorthosite-diorite; LJ_N, LJ_R, Lac St-Jean anorthosite (Buchan *et al.* 1983); Wi, Wilberforce pyroxenite; Co_A, Co_B, Co_C, Cordova gabbro (this study); In, Indian Head anorthosite; Fr, Frontenac Axis dykes; U1, U2, Umfraville gabbro; GD, Grenville dykes; SU, St Urbain anorthosite (Roy & Robertson 1979); GF, Grenville Front anorthosites; SM, Steel Mountain anorthosite (Murthy & Rao 1976); Tu, Tu_A, Tudor gabbro (Palmer & Carmichael 1973; Dunlop & Schutts 1983; Dunlop *et al.* 1985).

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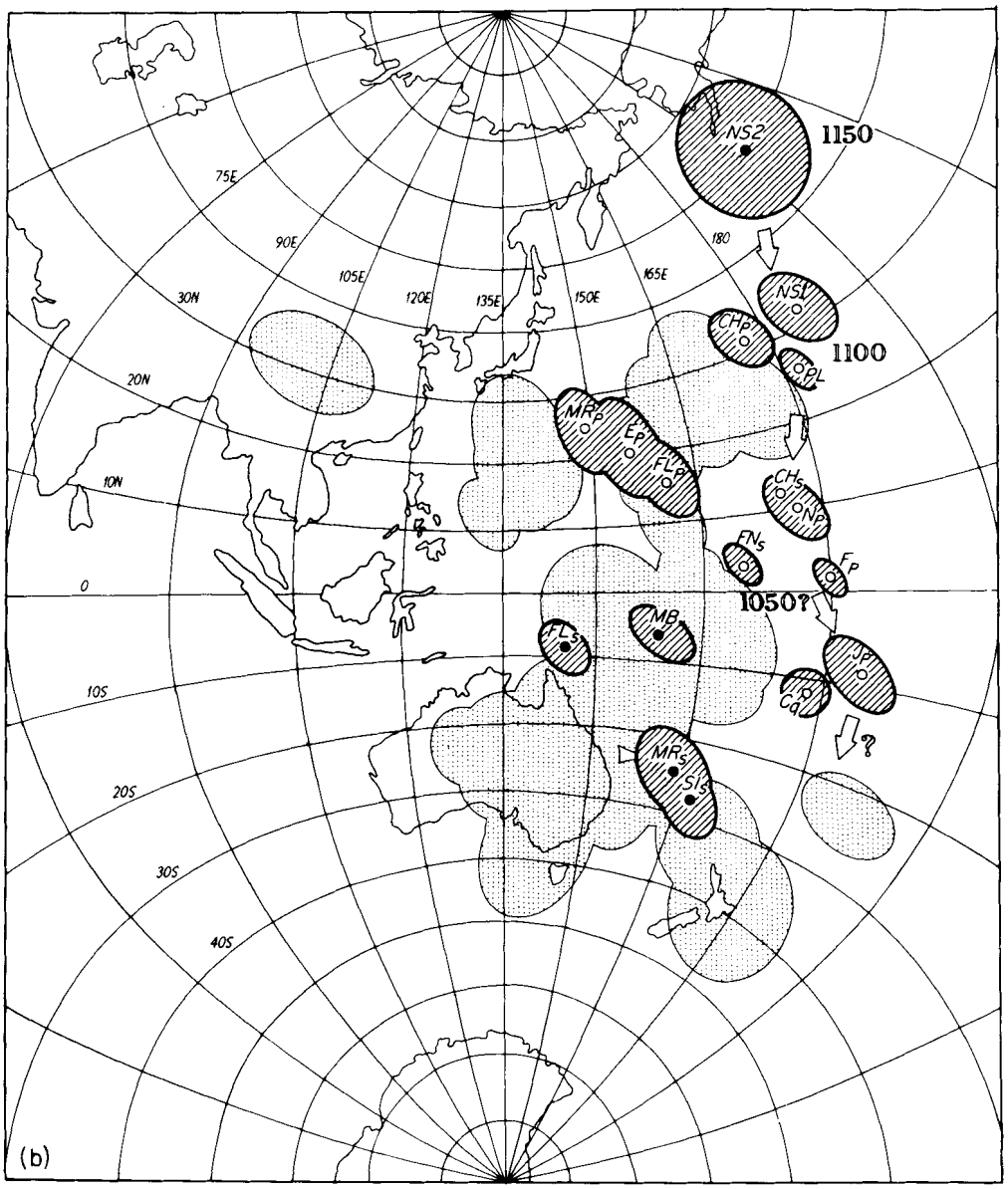


Figure 17. (b) Selected Keweenawan poles, referenced by Halls & Pesonen (1982, appendix 1) except as noted. The Grenville polar track of (a) is shown stippled, here and in (c) and (d). The polarity designation is compatible with (a). Key: NS1, NS2, North Shore volcanics; CH_P, CH_S, Copper Harbor volcanics; PL, Portage Lake volcanics; N_P, F_P, FN_S, Nonesuch shale primary, Freda sandstone primary, Freda and Nonesuch secondary; J_P, Jacobsville sandstone primary; MR_P, MR_S, Middle River section; E_P, Eileen section primary; FL_P, FL_S, Fond du Lac Formation; SI_S, Slate Islands secondary; MB, Michigan Borehole igneous (Van der Voo & Watts 1978); Cq, Chequamegon sandstone (McCabe & Van der Voo 1983).

TRMs acquired in this cooling, they would have to have characteristic unblocking temperatures of about 250 and $< 200^{\circ}C$ respectively. Such low-temperature TRMs would not have survived viscous overprinting by the PEF.

The A and B NRMs that we observe have overlapping laboratory unblocking temperature

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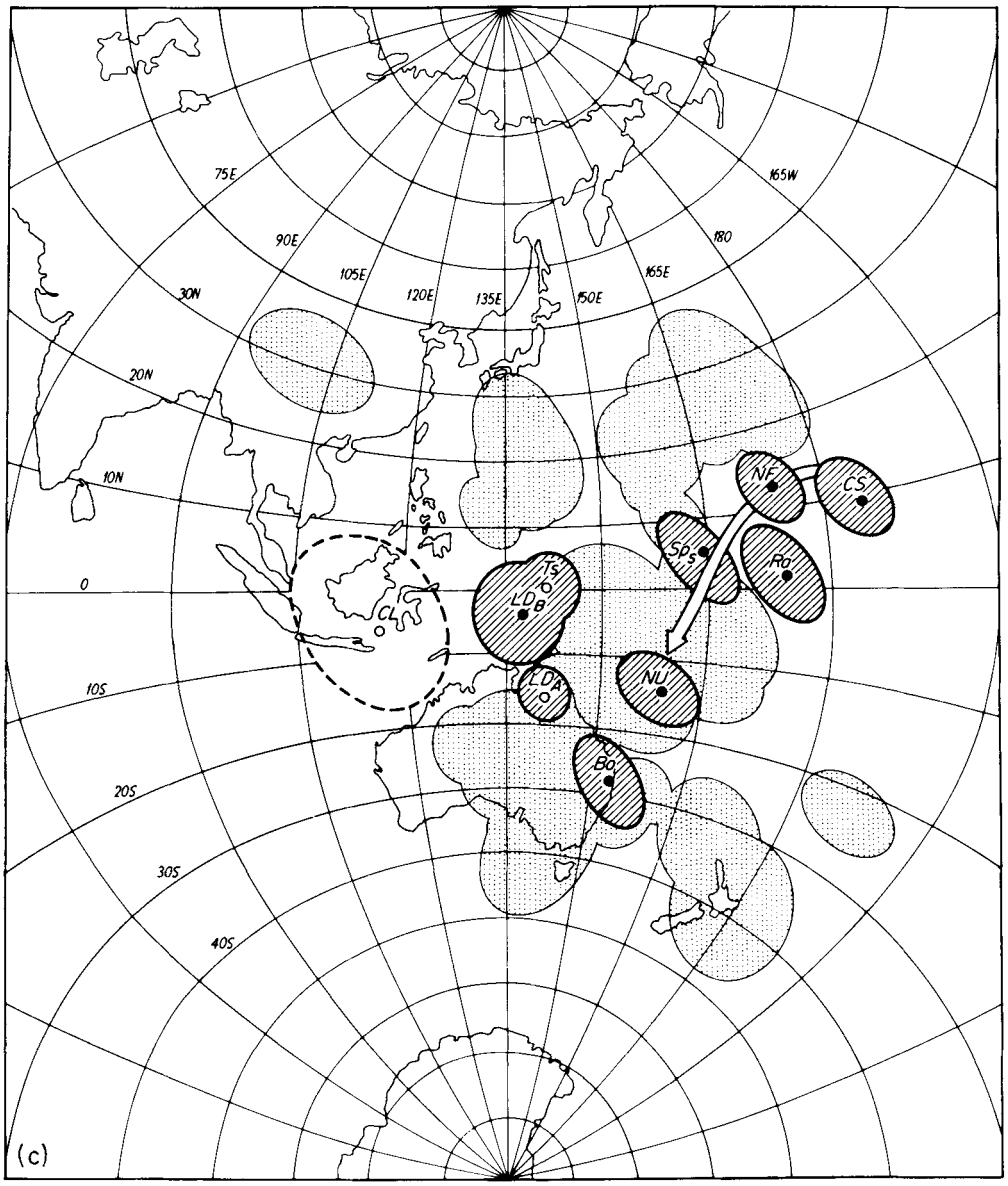


Figure 17. (c) Grenville-type poles from outside the Grenville Province, the Grenville Front region or the Keweenawan Basin. The first four poles are referenced in table 2 of McWilliams & Dunlop (1978). Radiometric ages of the rocks (not necessarily of the poles) are noted where known. Key: CS, NF, NU, Cardenas lavas, Nankoweap Formation ferrugineous and upper members, all of the Grand Canyon Supergroup; Ra, Rama diabase, Arizona, K/Ar 850–935 Ma; Sp_S, Spokane Formation secondary, Montana (Vitorello & Van der Voo 1977); Bo, Borden dykes, Baffin Island (Christie & Fahrig 1983); LD_A, LD_B, Ts, Little Dal red beds and Tsezotene sills, Mackenzie Mountains, NWT (Park 1981a, b); CL, Christmas Lake dyke, Wyoming, K/Ar and Rb/Sr 800–900 Ma (McClelland-Brown 1980).

spectra typically concentrated between 450 and 580°C (Fig. 6). In view of the low degree of alteration observed in A-component thin sections, A may well be a secondary TRM acquired in a brief thermal pulse to > 580°C around 900 Ma. Because of its overlapping unblocking temperatures, B must be in large part a CRM acquired around 850–800 Ma,

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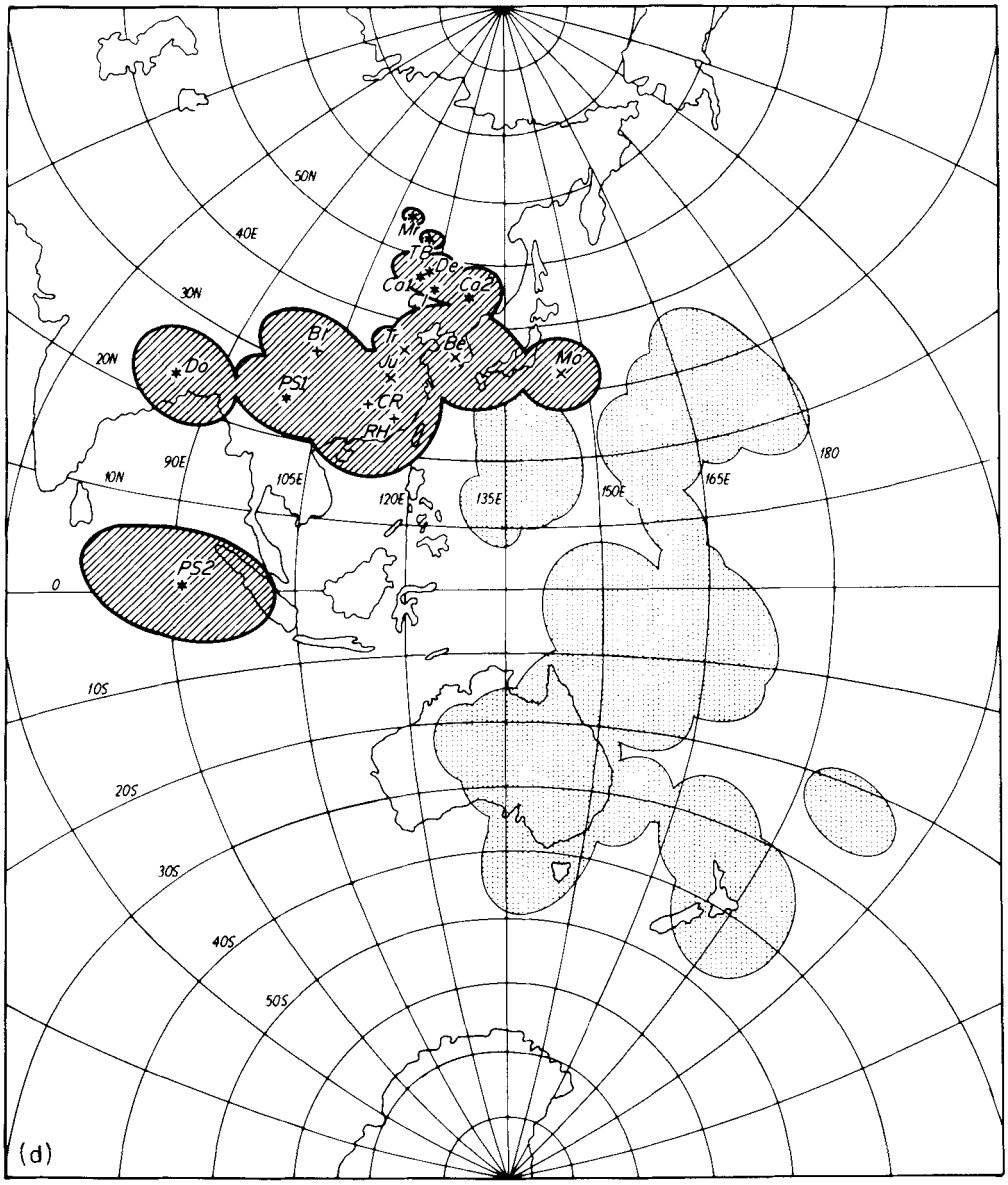


Figure 17. (d) Palaeopoles for middle Ordovician to upper Devonian rocks from cratonic North America as selected by Van der Voo *et al.* (1980). All but two of the poles are referenced there or in Buchan & Hodych (1982, table 3). Pole Co_C of the present study coincides almost exactly with pole Bl and its error ellipse. Rock ages are designated as middle or upper Ordovician (Om, Ou), lower, middle or upper Silurian (Sl, Sm, Su) or lower, middle or upper Devonian (Dl, Dm, Du), but this should not be taken to mean that the poles are necessarily primary. Key: Mo, Moccasin-Bays Formations, Om; Tr, Trenton limestone, Om-u; Ju, Juniata Formation, Ou; Be, Beemersville alkaline complex, Ou-Sl; CR, Chapman Ridge Formation, O-Sl; RH, Rose Hill Formation, Sm; Bl, Bloomsburg Formation, Su; PS1, PS2, Peel Sound Formation, Pl (Dankers 1982); Do, Dockendorff Group, D; Cl, Columbus limestone, DM; De, Delaware limestone, Dm; Mr, Martin Formation, Dm-u; TB, Temple Butte limestone, Dm-u; Ca1, Ca2, Catskill Formation, Dm-u.

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either below 250°C during slow continuous cooling or below about 400°C in a brief thermal pulse.

The cause of these postulated discrete NRM overprinting events is unknown, but the suggestion is clear that there may be unresolved 'fine structure' in present Grenvillian cooling curves. Brief events or pulses around 900 and 850–800 Ma, superimposed on general uplift and cooling, would account for the marked clustering of palaeopoles around these times on the Grenville Track (Fig. 17a). It has long been puzzling that Grenvillian magnetizations with broad ranges of laboratory blocking temperatures should fail to record APW if they were indeed acquired during continuous slow cooling. They are typically resultants of discrete A and B remanences, not continuously evolving vectors that track the changing Earth's field direction during slow cooling. Vectors of the latter type are well documented in slowly cooled rocks from west Greenland (Morgan 1976; Piper 1981).

It can be argued that the very segregation of data into A and B groups, like those of Figs 4 and 8, suppresses any between-site record of APW, and authors have occasionally plotted site-mean poles to make the point that they span much of the Grenville Track (see, e.g. Roy & Robertson 1979, fig. 1). However, there is no escaping the fact that discrete A and B remanences are resolved quite routinely in orthogonal vector plots of individual specimen data (e.g. Fig. 9). Discrete pulses superimposed on general Grenvillian cooling would account for this observation.

THE C NRM: PALAEOZOIC REACTIVATION IN THE GRENVILLE

The C magnetization tends to occur in samples with pronounced silicate alteration and its blocking temperatures are lower on the average than those of A and B NRMs (Fig. 6), although there is considerable overlap among A, B and C spectra. On this evidence, C is a low-temperature CRM younger than either A or B. Despite its low AF coercivities and its association with samples containing abundant visible opaques of multidomain size, the C component is probably carried mainly by very fine, nearly superparamagnetic secondary magnetite. Only such fine-grained material has low blocking temperatures as well as low coercivities.

Results from the Cordova Mines cores do not support a genetic relation between early Palaeozoic mineralization and the C magnetization, although they do not rule out such a relation. The C component is certainly no more widespread in these cores than in the general sampling.

There was definitely a late Ordovician or early Silurian (about 445 Ma) event that outgassed argon from plagioclases in the Cordova samples (Lopez-Martinez & York 1983). It is plausible that the temperatures reached did not exceed 200°C or so, and that the C magnetization was produced thermochemically in this event. The good agreement of the Co_C palaeopole with known Silurian and late Ordovician poles for cratonic North America makes such an origin almost certain.

The timing of this event suggests an echo of the Taconic orogeny in the Appalachian region. If so, the relation between the two is subtle. Middle Ordovician sediments just south of Crowe Lake (Fig. 2) and for at least 100 km to the south and east are flat lying and undisturbed. Alternatively, mild reheating due to burial by these sediments, which must once have overlain the Cordova body, or alteration related to seawater infiltration during sediment deposition could have caused the C overprinting (D. T. A. Symons 1984, private communication).

Whatever the exact nature of the C magnetization, if one accepts that it is a late overprint and that its age is fixed beyond reasonable doubt by the $^{40}Ar/^{39}Ar$ plagioclase date of

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445 Ma, then palaeopole Co_C takes on a pivotal role. The dating of most North American Silurian and Ordovician poles remains rather uncertain. There are few radiometric ages and the possibility remains that these poles, which form a broad band (Fig. 17d) near both Permian and Carboniferous poles, could record pervasive later Palaeozoic remagnetization (see, e.g. discussion in Roy, Tanczyk & Lapointe 1983).

Co_C *is* an overprint, but an Ordovician or Silurian one, and it comes from a formation well outside the Appalachian orogen. Its pole position is strong evidence that middle Ordovician to Silurian poles from Appalachian rocks are primary, not secondary.

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